THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 5: **Arithmetic Progressions** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The common difference of the AP $3, 1, -1, -3, \ldots$ is: 1

(a) 2 **(b)** −2 **(c)** 3 **(d)** −3

2. Which of the following is an AP? 1

(a) $2, 4, 8, 16, \ldots$ **(b)** $1, 3, 9, 27, \ldots$ **(c)** $-10, -6, -2, 2, \ldots$ **(d)** $1, 1, 2, 3, \ldots$

3. The 11th term of the AP $-3, -\frac{1}{2}, 2, \ldots$ is: 1

(a) 22 **(b)** 28 **(c)** −38 **(d)** $-48\frac{1}{2}$

4. If $2x,\ x+10,\ 3x+2$ are in AP, then x is: 1

(a) 6 **(b)** 4 **(c)** 3 **(d)** 2

5. A student states: "The sum of the first n odd natural numbers is $n(n+1)$." The correct response is: 1

(a) Correct **(b)** Incorrect; the sum is n^2

(c) Correct only for even n **(d)** Cannot be determined

6. The nth term of the AP $7, 11, 15, \ldots$ is: 1

(a) $4n+3$ **(b)** $4n+7$ **(c)** $7n+4$ **(d)** $3n+4$

7. For the AP $5, 9, 13, \ldots$, Student A finds $a_{20} = 81$ and Student B finds $a_{20} = 85$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. The sum of the first 10 terms of $2, 4, 6, \ldots$ is: 1

(a) 100 **(b)** 110 **(c)** 120 **(d)** 90

9. The number of two-digit numbers divisible by 3 is: 1

(a) 30 **(b)** 31 **(c)** 32 **(d)** 33

10. If $S_n = n^2 + 2n$, the common difference of the AP is: 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Which term of the AP $3, 8, 13, 18, \ldots$ is 78? 2

12. Find the sum: $7 + 10\frac{1}{2} + 14 + \cdots + 84$. 2

13. Check whether 301 is a term of the AP $5, 11, 17, 23, \ldots$ 2

14. A student says that the 30th term of the AP $10, 7, 4, \ldots$ is -77. Verify. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the sum of the first 15 multiples of 8. 3

16. The 3rd and the 9th terms of an AP are 4 and -8 respectively. Which term of this AP is zero? 3

17. How many terms of the AP $24, 21, 18, \ldots$ must be taken so that their sum is 78? Explain the double answer. 3

OR

The sum of the 4th and 8th terms of an AP is 24 and the sum of the 6th and 10th terms is 44. Find the first three terms of the AP. 3

18. A contract on a construction job specifies a penalty for delay: ₹200 for the first day, ₹250 for the second day, ₹300 for the third day, and so on. How much money does the contractor pay as penalty if he delays the work by 30 days? 3

OR

Ramkali saved ₹5 in the first week of a year and then increased her weekly savings by ₹1.75. If in the nth week her weekly saving becomes ₹20.75, find n. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Subba Rao started work in 1995 at an annual salary of ₹5000 and received an increment of ₹200 each year. 5

(i) In which year did his income reach ₹7000?

(ii) Find his total earnings in the first 11 years.

(iii) A student claims his salary in the 20th year was ₹9000. Justify or refute.

OR

200 logs are stacked such that there are 20 logs in the bottom row, 19 in the next row, 18 in the row next to it, and so on. In how many rows are the 200 logs placed and how many logs are in the top row? 5

20. The sum of the first 7 terms of an AP is 49 and that of the first 17 terms is 289. Find the sum of the first n terms. Verify your result for $n = 10$. 5

OR

A spiral is made up of successive semicircles, with centres alternately at A and B, starting with centre at A, of radii 0.5 cm, 1.0 cm, 1.5 cm, 2.0 cm, … What is the total length of such a spiral made up of thirteen consecutive semicircles? (Take $\pi = \frac{22}{7}$) 5

END OF QUESTION PAPER — **Total: 40 Marks**